ISEG Module Qualification Standard

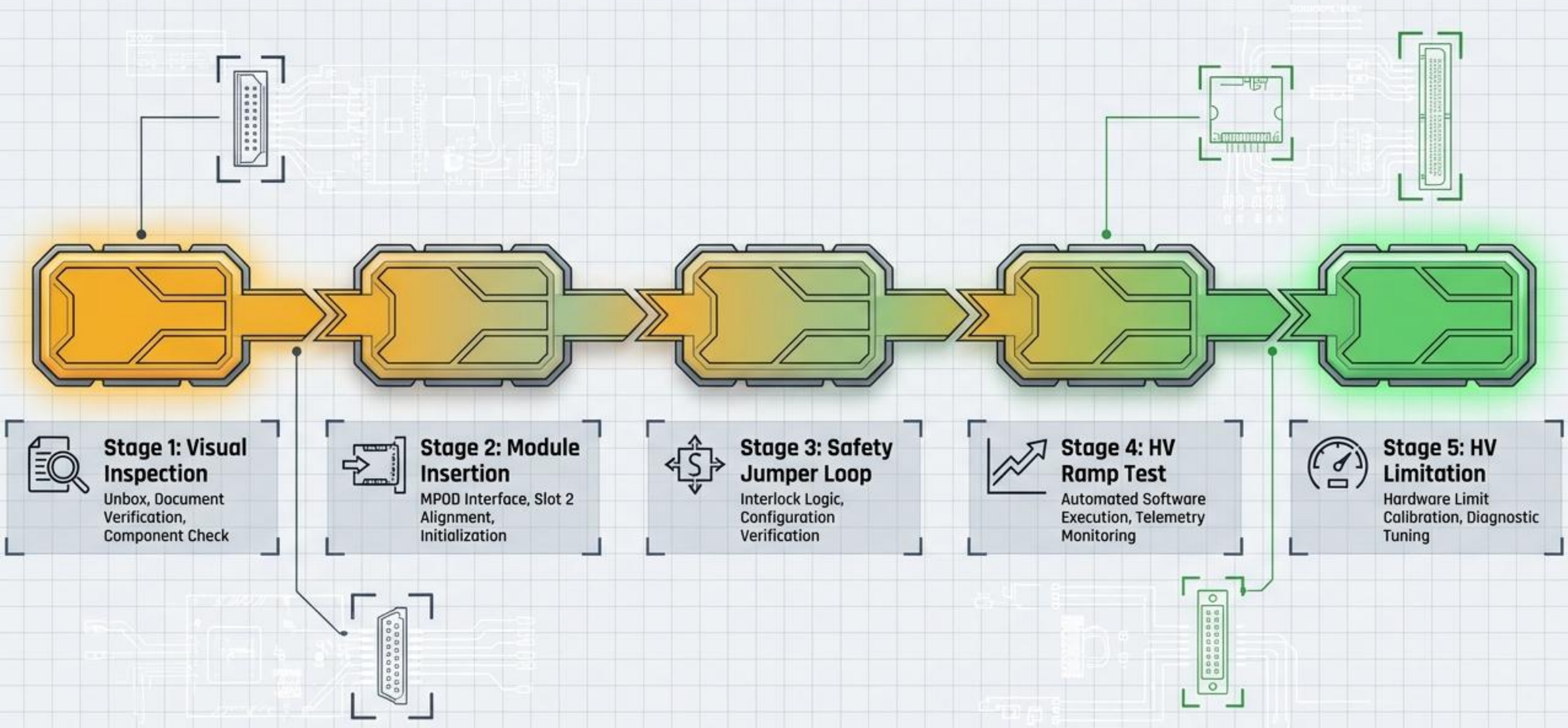
The 5-Stage DUNE WIEC High-Voltage Test Flow

Standard Operating Procedure for Lab Technicians and QA Engineers.



Gemini Notebook

The Verification Gauntlet



Stage 1: Visual Inspection

Unbox, Document Verification, Component Check

Stage 2: Module Insertion

MPOD Interface, Slot 2 Alignment, Initialization

Stage 3: Safety Jumper Loop

Interlock Logic, Configuration Verification

Stage 4: HV Ramp Test

Automated Software Execution, Telemetry Monitoring

Stage 5: HV Limitation

Hardware Limit Calibration, Diagnostic Tuning

Gemini Notebook

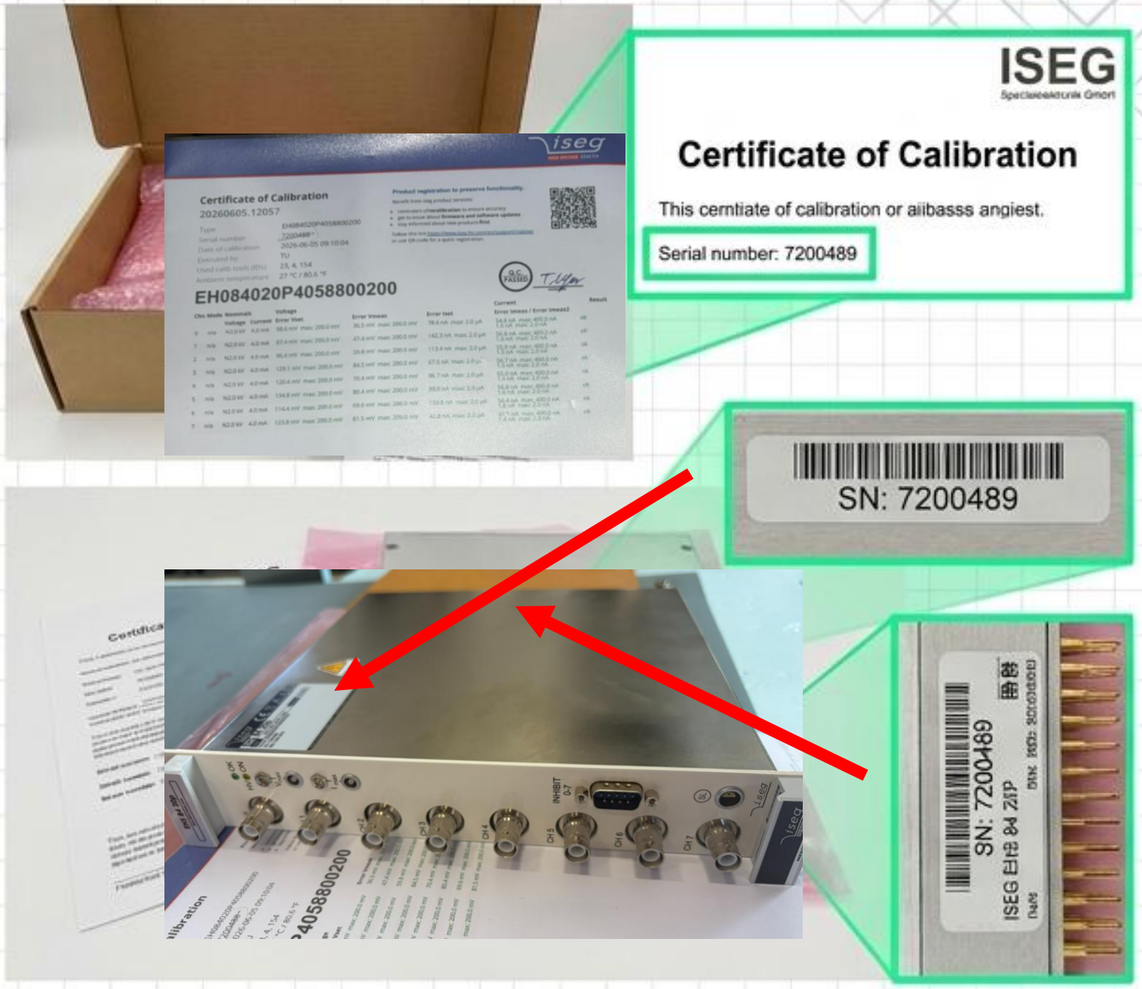
Initial Unboxing & Documentation Match

Checklist telemetry dashboard

- Extract Calibration Certificate from static-dissipative packaging.
- Locate hardware Serial Number (SN) on the ISEG module chassis.
- Verify SN exactly matches the Certificate of Calibration and box label.

Action Required: If serial numbers do not align, abort procedure and quarantine module.

Expected vs. Actual



Gemini Notebook

Hardware Integrity Check

Front Panel Inspection

Verify all primary connectors are seated firmly and free of physical damage. Check that blanking plates or adjacent slots are clear.



Backplane Inspection

⚠ **Safety Pre-Check:**
Confirm the default Safety Jumper (SL-disable) is physically present in its designated slot prior to first insertion.



Inspect the multi-pin connector for bent or recessed pins.

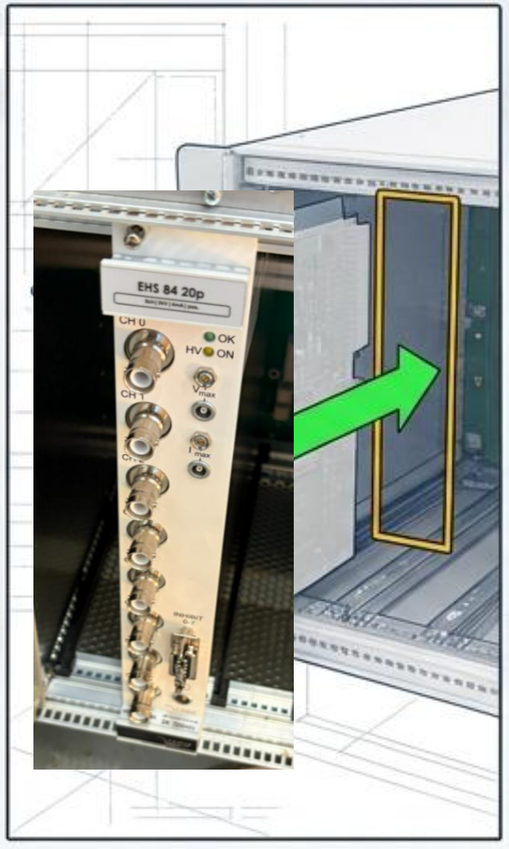


Gemini Notebook

Crate De-Energization & Module Seating

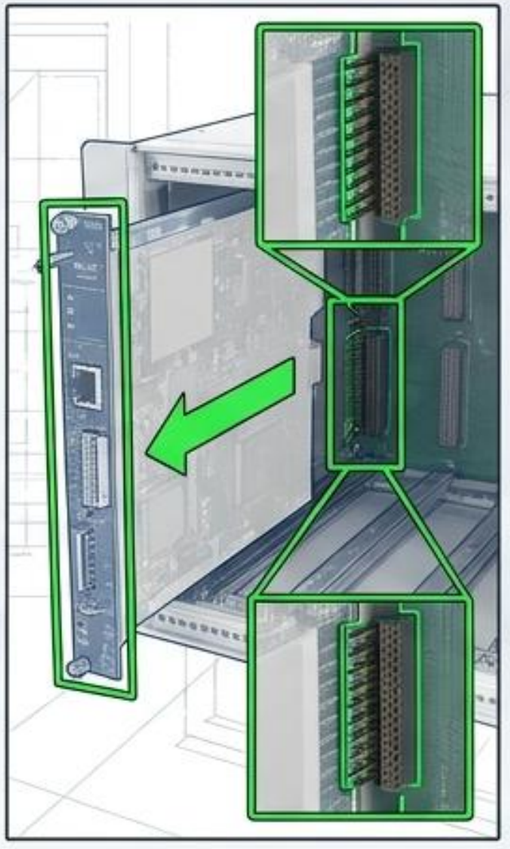
Critical Safety Gate: MPOD crate power MUST be fully toggled before any hardware manipulation.

ALERT: Failure to de-energize may result in equipment damage or personal injury. Confirm zero voltage before proceeding.



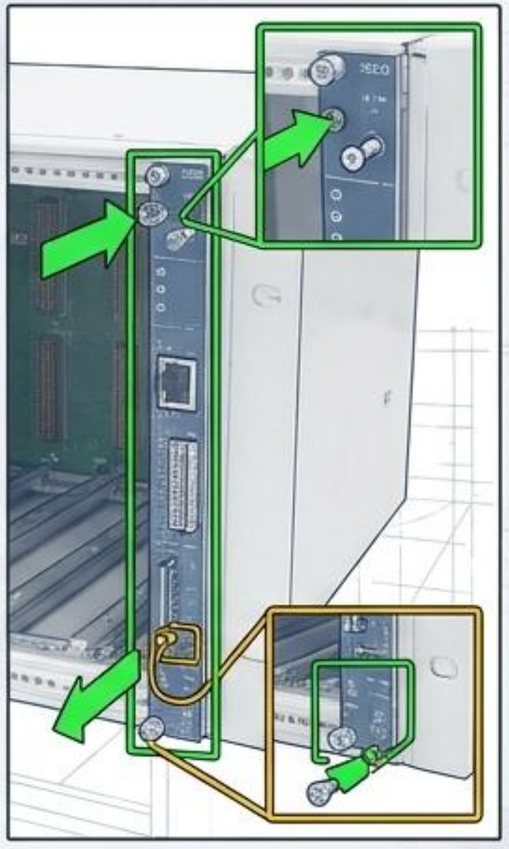
Step 1: Align

Align the ISEG module with Slot 2 of the MPOD crate.



Step 2: Slide

Slide the module smoothly along the guide rails until the backplane pins engage fully.



Step 3: Secure

Secure the module faceplate to the crate chassis to prevent accidental unseating during subsequent jumper tests.

Gemini Notebook

Action: Restore power to the MPOD crate.

System Initialization & Handshake

Hardware Verification



Locate the top indicator LEDs on the ISEG front panel. The OK light **must** illuminate solid Neon Green.

Software Verification

```
MPOD_TERM> INITIATE_INTERFACE
[INFO] Establishing connection...
[INFO] Connection successful.

[CMD] READ_SN SLOT 2
[INFO] Module detected in Slot 2.
[RESULT] Serial Number: ISEG-EHS-20P-SN-00145
[INFO] Verification Complete. Module Online.
```

Initiate the MPOD terminal interface. The system must successfully read and report the module's Serial Number in Slot 2.

Status: If Green LED is active and SN reads out accurattly, the module is ready for interlock testing.

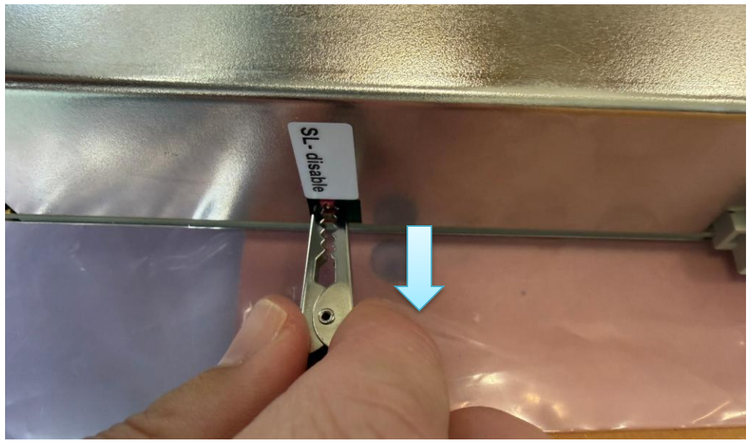
Gemini Notebook

Safety Test 01: Remove Crate Module and Jumper





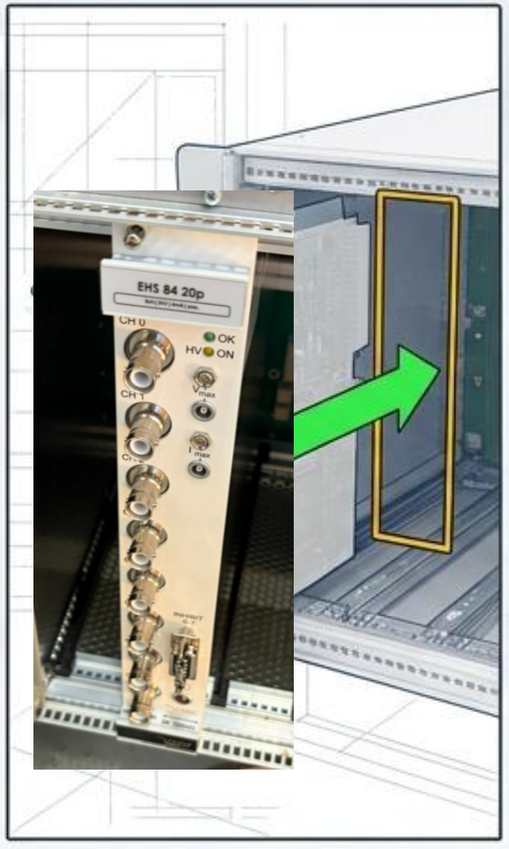




Crate De-Energization & Module Seating

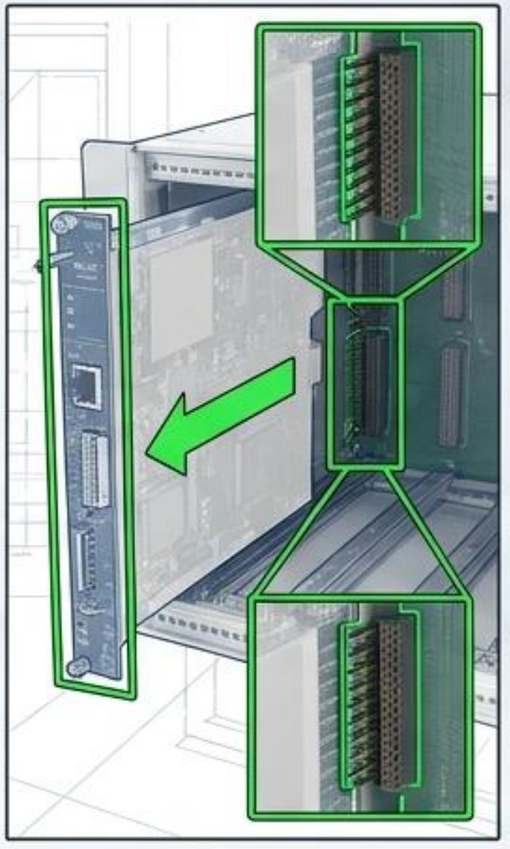
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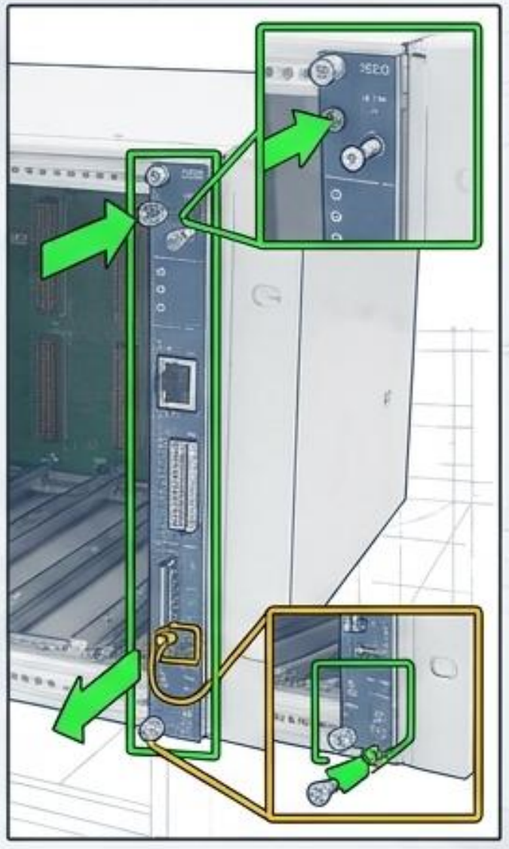
Step 1: Align

Align the ISEG module with Slot 2 of the MPOD crate.



Step 2: Slide

Slide the module smoothly along the guide rails until the backplane pins engage fully.



Step 3: Secure

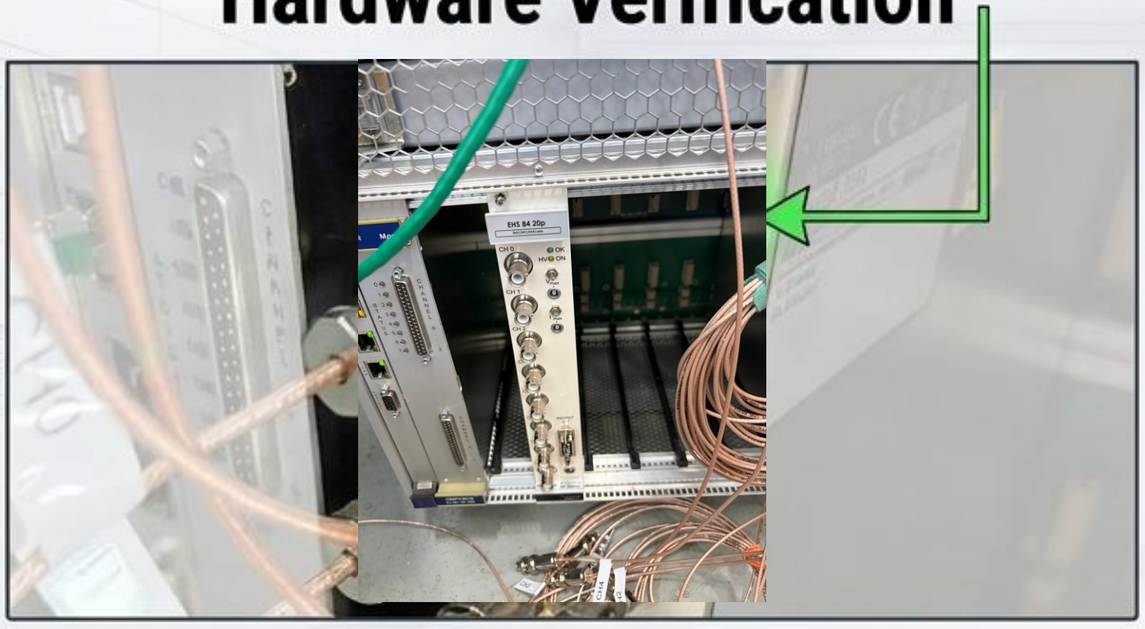
Secure the module faceplate to the crate chassis to prevent accidental unseating during subsequent jumper tests.

Gemini Notebook

Action: Restore power to the MPOD crate.

System Initialization & Handshake

Hardware Verification



At this condition, the LED is off

Software Verification

```
MPOD_TERM> INITIATE_INTERFACE
[INFO] Establishing connection...
[INFO] Connection successful.

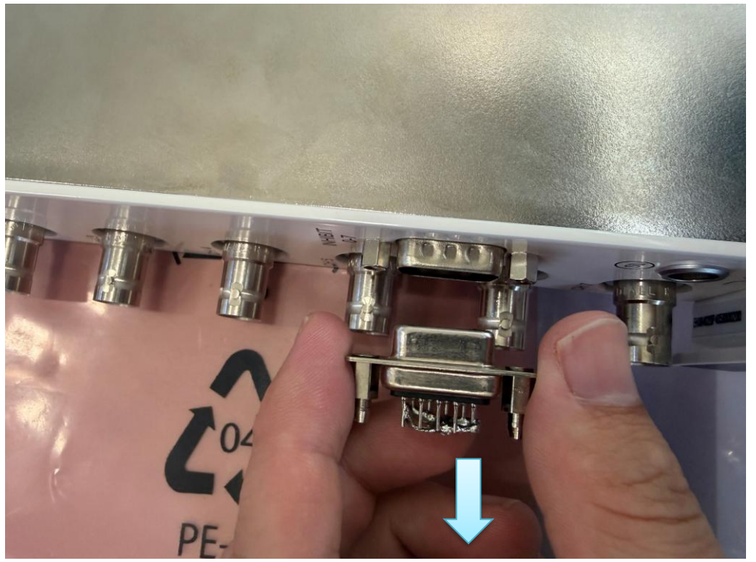
[CMD] READ_SN SLOT 2
[INFO] Module detected in Slot 2.
[RESULT] Serial Number: ISEG-EHS-20P-SN-00145
[INFO] Verification Complete. Module Online.
```

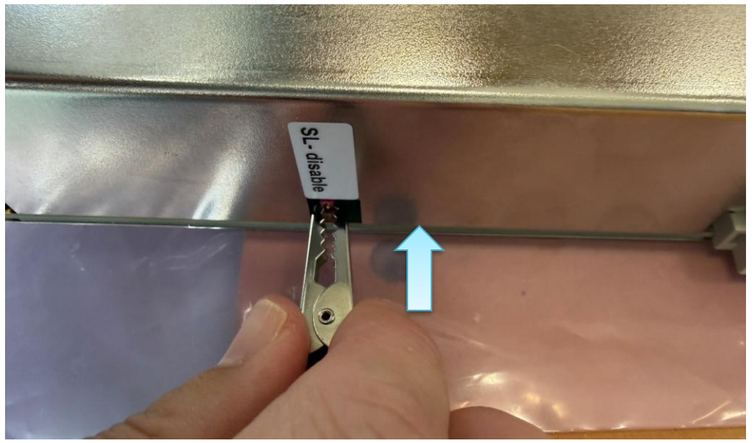
Initiate the MPOD terminal interface. The system must successfully read and report the module's Serial Number in Slot 2.

Safety Test 02: Remove Crate Module and test components





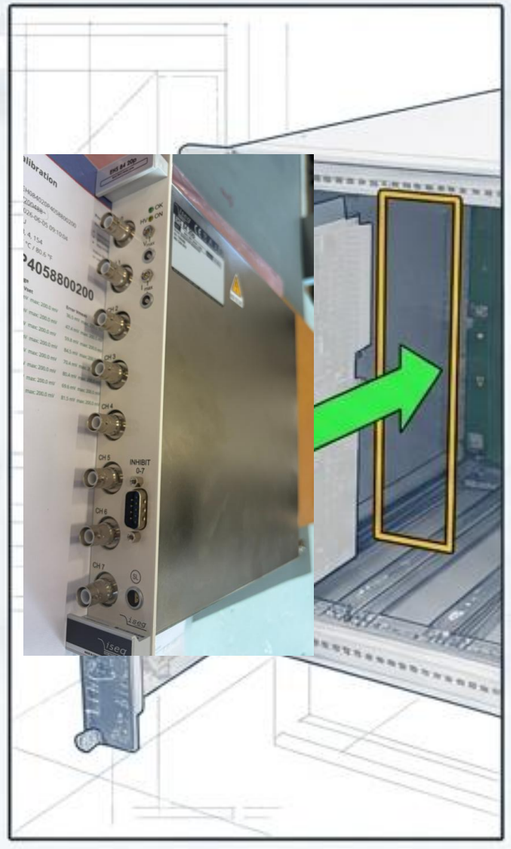




Crate De-Energization & Module Seating

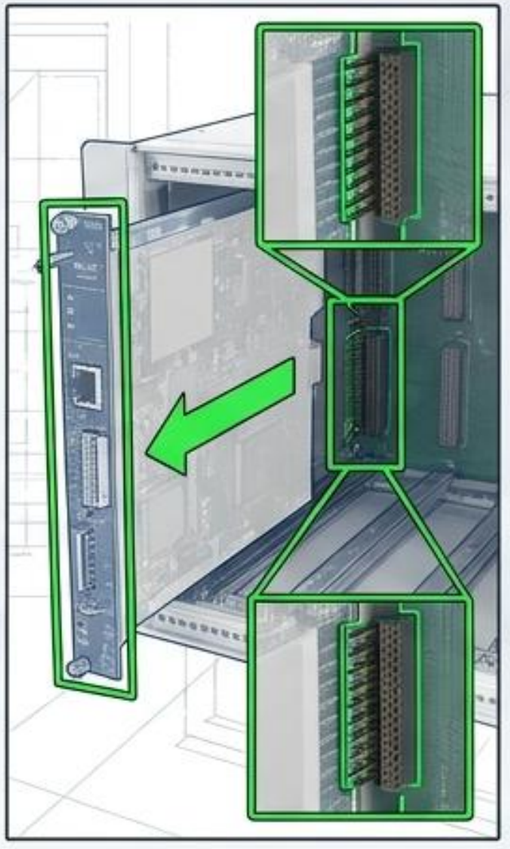
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ALERT: Failure to de-energize may result in equipment damage or personal injury. Confirm zero voltage before proceeding.



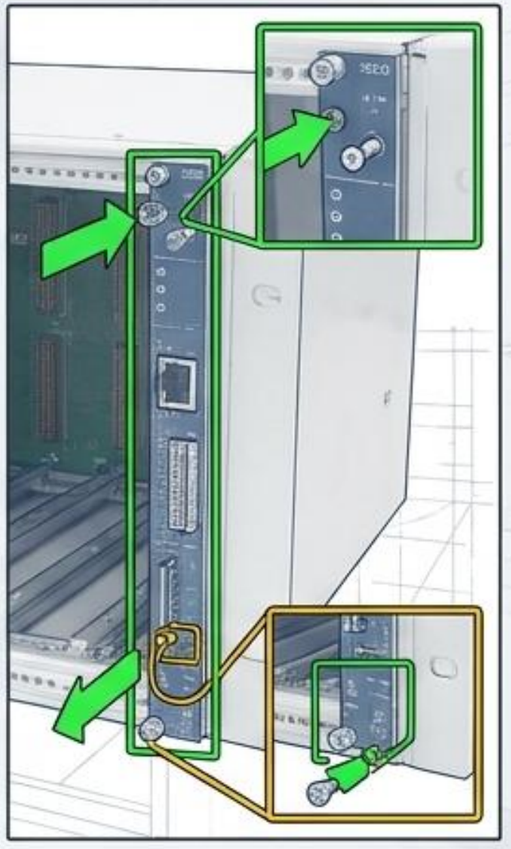
Step 1: Align

Align the ISEG module with Slot 2 of the MPOD crate.



Step 2: Slide

Slide the module smoothly along the guide rails until the backplane pins engage fully.



Step 3: Secure

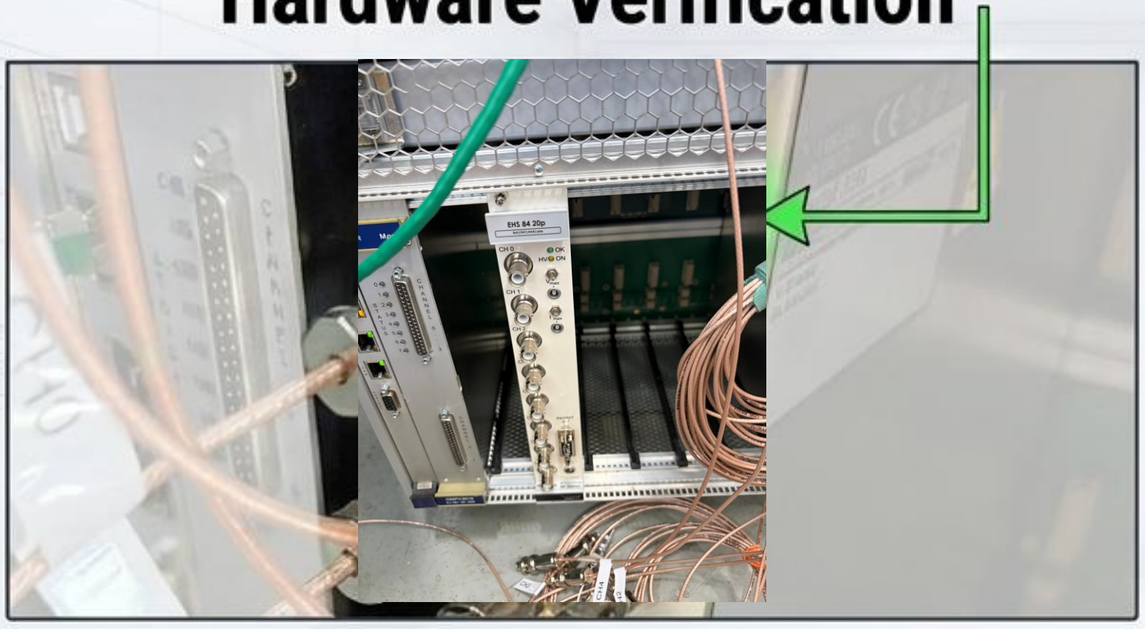
Secure the module faceplate to the crate chassis to prevent accidental unseating during subsequent jumper tests.

Gemini Notebook

Action: Restore power to the MPOD crate.

System Initialization & Handshake

Hardware Verification



At this condition, the LED is off

Software Verification

```
MPOD_TERM> INITIATE_INTERFACE
[INFO] Establishing connection...
[INFO] Connection successful.

[CMD] READ_SN SLOT 2
[INFO] Module detected in Slot 2.
[RESULT] Serial Number: ISEG-EHS-20P-SN-00145
[INFO] Verification Complete. Module Online.
```

Initiate the MPOD terminal interface. The system must successfully read and report the module's Serial Number in Slot 2.

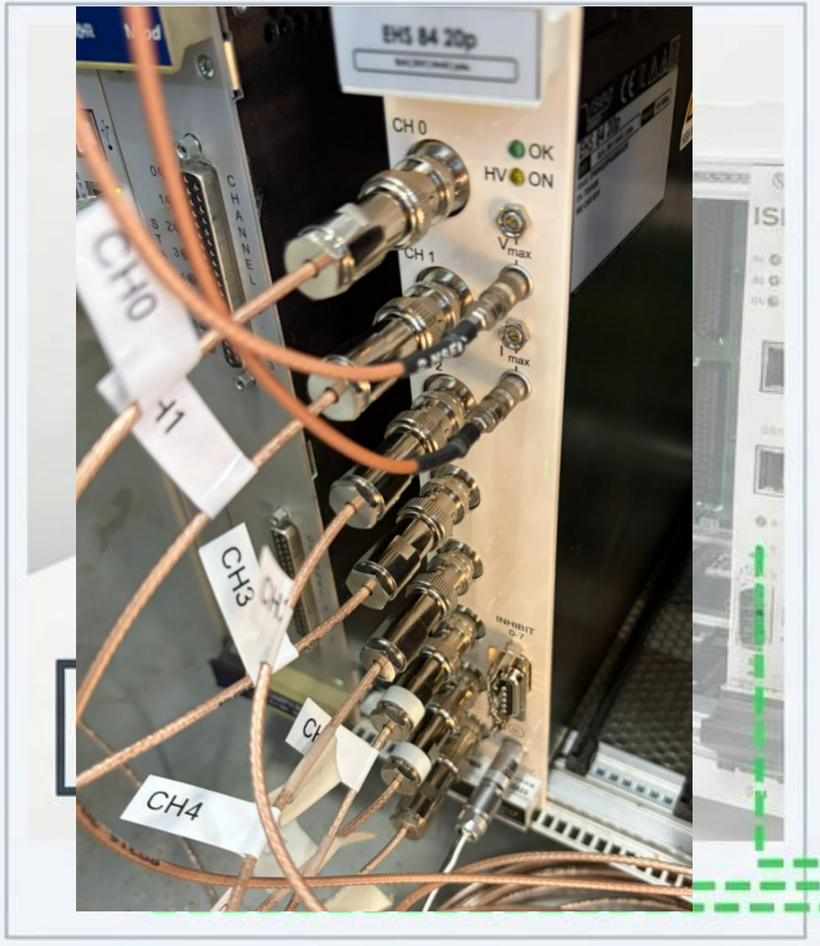
Restoring Normal Configuration & Arming

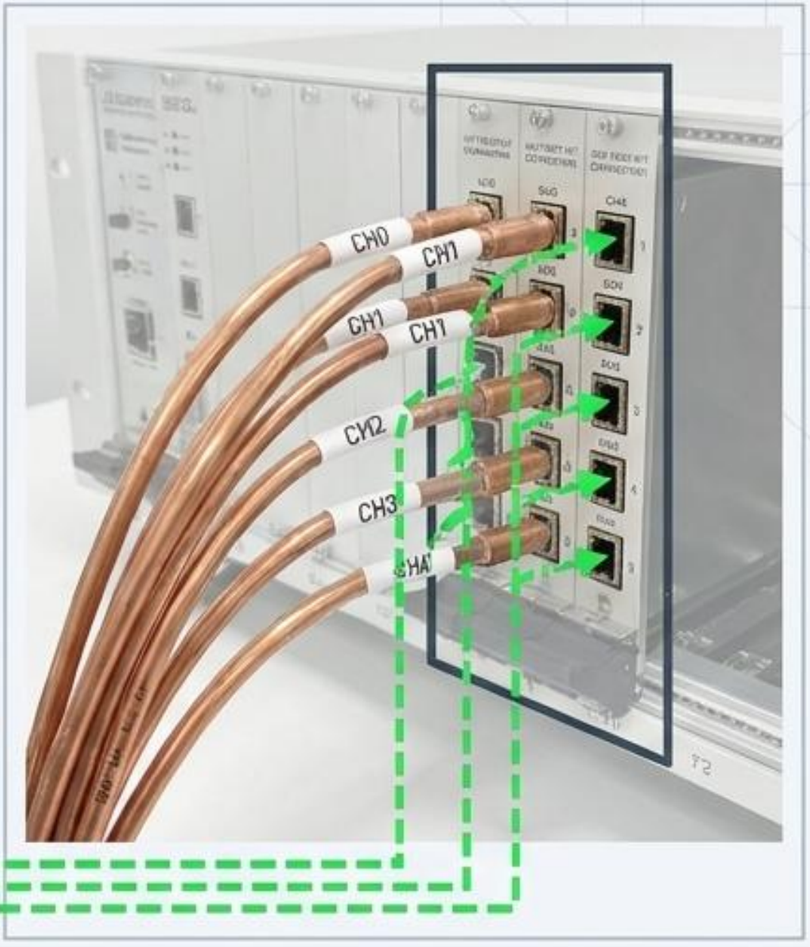
Step 1: Power down and extract module.

Step 2: Re-install the red SL-disable safety jumper into the backplane pin block. Verify it is fully seated.

Step 3: Re-insert module into Slot 2 and secure it.

Step 4: Install the dedicated HV Test Connectors onto the front panel channels.



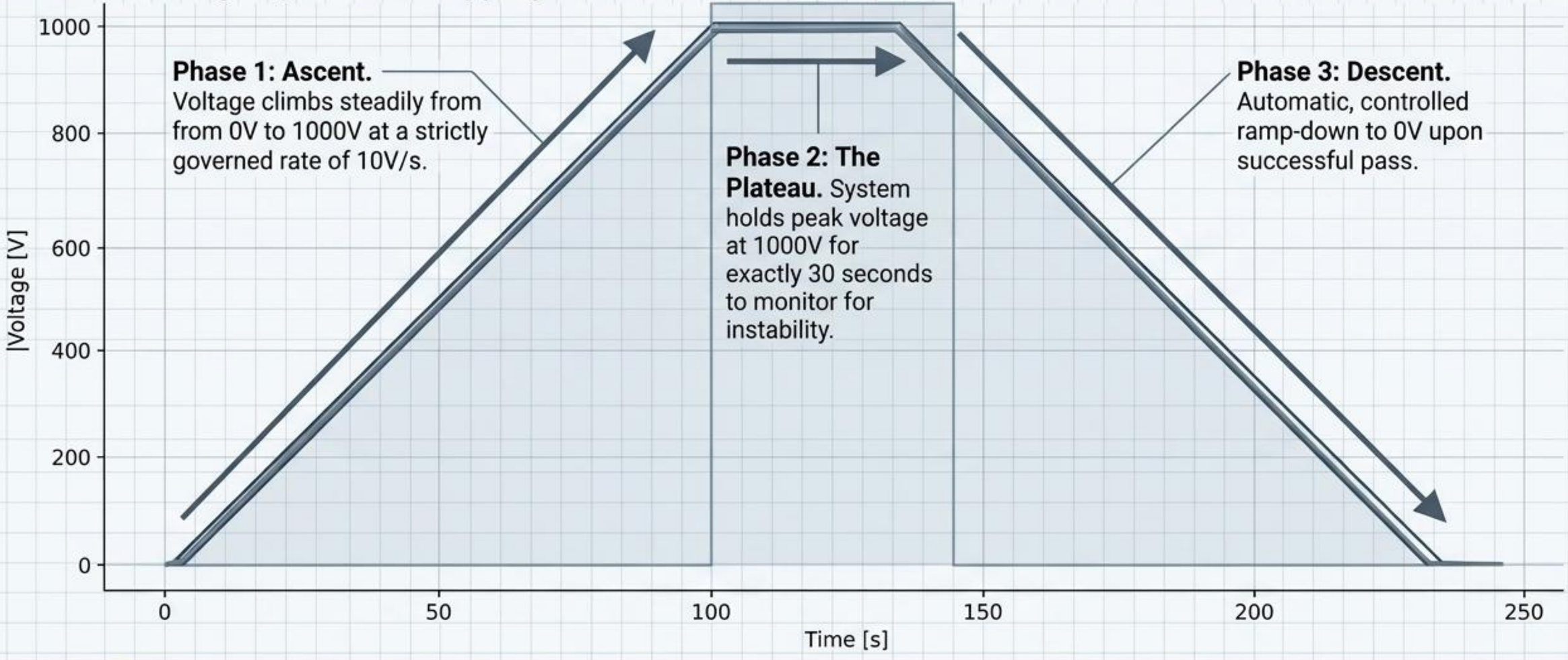


State Confirmation: System logic now reports the safety loop as closed. The module is armed for Stage 4 automated testing.

Gemini Notebook

The High-Voltage Ramp Flight Plan

Test Parameter Overview: An automated, software-driven stress test to verify smooth voltage scaling without arcing or protective tripping.



⚠ **Note:** Any protective trip during this cycle immediately aborts the test and grounds the system.

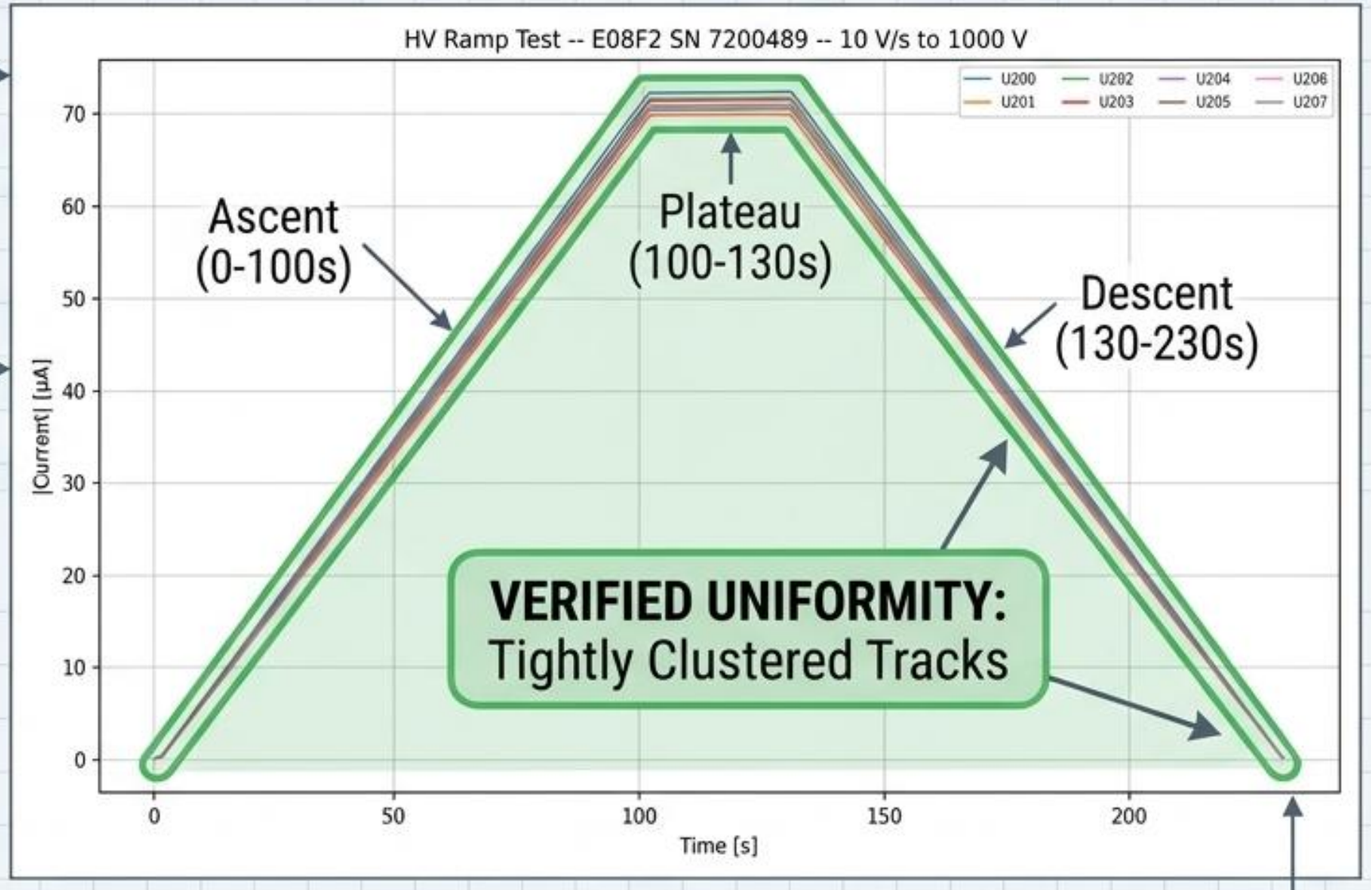
Gemini Notebook

Current Response Profile Verification

Diagnostic Focus: Current (μA) must scale linearly and symmetrically with the voltage ramp.

Verification Check: Observe the multi-channel current tracks. They must remain tightly clustered. ✅

Failure Mode: Significant divergence between channels or sudden, jagged spikes in the current profile indicate internal leakage or component failure, requiring immediate module rejection.



Gemini Notebook

Live Telemetry & Automated Safeguards

Action: Execute the Python automation script via the MPOD terminal.

Monitoring: Watch the real-time console output.

```
15:49:25 Vmax    1.562479 V   target 1.5625 V   diff -0.000021 V   IN RANGE
  Vmax is good now -- press Enter to accept
15:49:26 Vmax    1.562479 V   target 1.5625 V   diff -0.000021 V   IN RANGE
  Vmax is good now -- press Enter to accept
15:49:27 Vmax    1.562466 V   target 1.5625 V   diff -0.000034 V   IN RANGE
  Vmax is good now -- press Enter to accept
15:49:27 Vmax    1.562466 V   target 1.5625 V   diff -0.000034 V   IN RANGE
  Vmax is good now -- press Enter to accept
15:49:28 Vmax    1.562453 V   target 1.5625 V   diff -0.000047 V   IN RANGE
  Vmax is good now -- press Enter to accept
15:49:29 Vmax    1.562466 V   target 1.5625 V   diff -0.000034 V   IN RANGE
  Vmax is good now -- press Enter to accept

15:48:55 Vmax    1.563260 V   target 1.5625 V   diff +0.000760 V   turn DOWN
15:48:56 Vmax    1.551580 V   target 1.5625 V   diff -0.001920 V   turn UP
15:48:57 Vmax    1.552528 V   target 1.5625 V   diff -0.009972 V   turn UP
15:48:57 Vmax    1.552420 V   target 1.5625 V   diff -0.010010 V   turn UP
15:48:58 Vmax    1.552515 V   target 1.5625 V   diff -0.009985 V   turn UP
15:48:59 Vmax    1.554590 V   target 1.5625 V   diff -0.007910 V   turn UP
15:49:00 Vmax    1.557958 V   target 1.5625 V   diff -0.004542 V   turn UP
15:49:00 Vmax    1.557651 V   target 1.5625 V   diff -0.004449 V   turn UP
15:49:01 Vmax    1.560878 V   target 1.5625 V   diff -0.001058 V   turn UP
15:49:02 Vmax    1.561422 V   target 1.5625 V   diff -0.001174 V   turn UP
```

Target Identification: The system continuously calculates the difference between V_{max} and target.

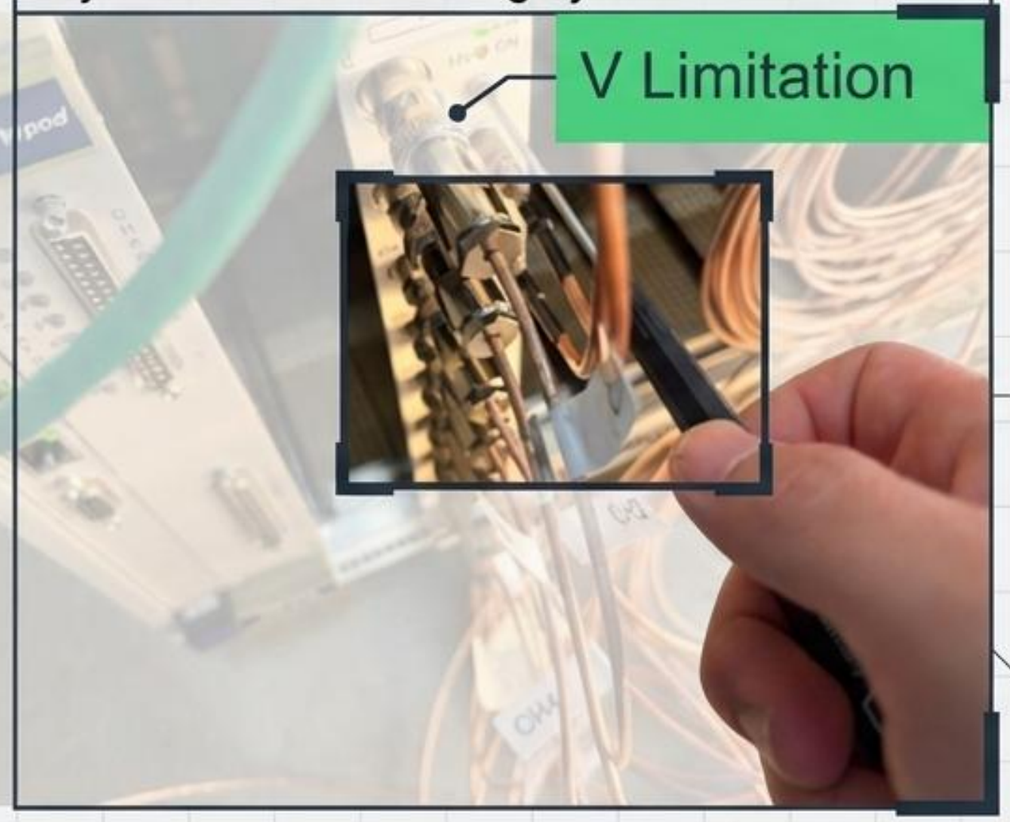
Success State: The terminal will repeatedly confirm diff is negligible and print IN RANGE. The operator presses Enter only when prompted by the software to accept the good state.

Gemini Notebook

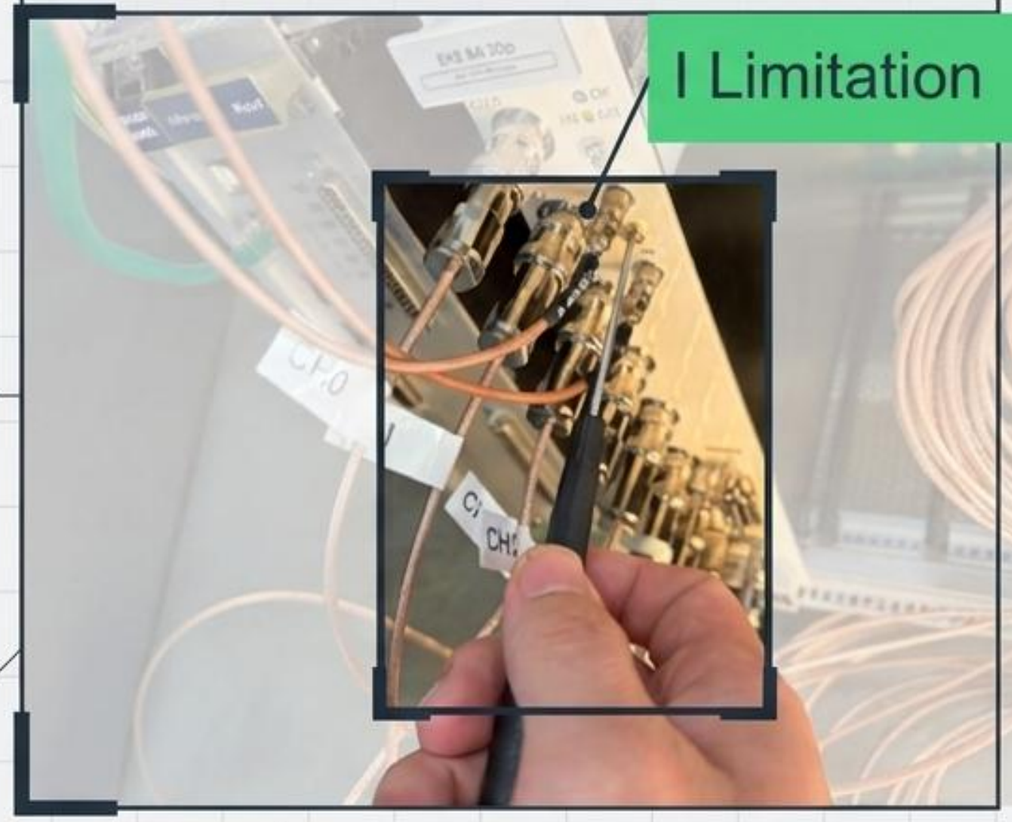
Physical Limitation Calibration Setup

Objective: Validate the hardware safety limits (Vmax and Imax) against control voltage inputs.
Tool Required: Precision flathead calibration screwdriver.

Action 1 (Voltage): Locate the V Limitation potentiometer on the front panel. Prepare to adjust while monitoring system readout.



Action 2 (Current): Locate the I Limitation potentiometer immediately below it.



Gemini Notebook

The Limitation Translation Matrix

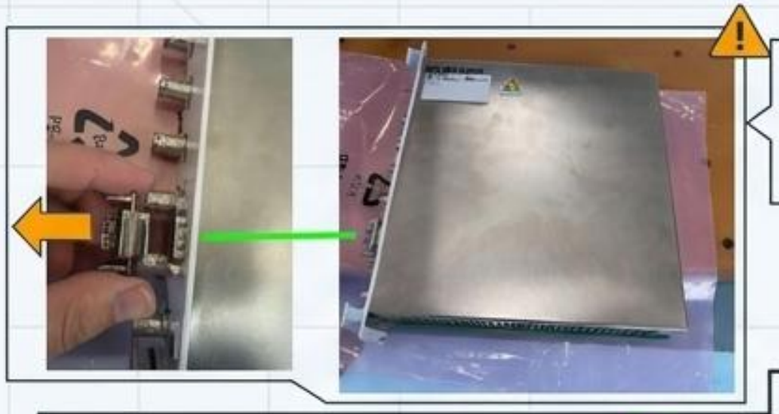
Instruction: Adjust the front-panel potentiometers to achieve the following specific system readouts. Do not exceed these bound limits.

Section 1: Voltage Limit (Vmax)		Section 2: Current Limit (Imax)	
Input **2.5V**	Hard Limit **2.0kV**	Input **2.5V**	Hard Limit **4.0mA**
Input **1.5625V**	Hard Limit **1.25kV**	Input **0.15625V**	Hard Limit **0.25 mA**

Note: The current limit exhibits a flat-top curve at lower voltages to protect the circuitry.

Gemini Notebook

Teardown & Safe Repackaging



Extract Module, Verify Safety Jumper





Seal & Certificate Verified



DUNE Deployment Ready

Step 3: Extract the ISEG module. Verify the safety jumper remains installed.

Step 2: Remove HV Test Connectors.

Step 1: Terminate software connection and completely power off the MPOD crate.

Step 4: Seal the module alongside its matched Calibration Certificate back into the static-dissipative bag.

Step 5: Box the unit securely with protective foam, ready for DUNE deployment.

Gemini Notebook

The Safety Loop Interlock Matrix

The interlock circuit prevents HV discharge if the safety loop is broken.
We must verify all three failure/run states.

		
State: Jumper Only (No Test Kit)	**State: Test Kit Only (Jumper Removed)**	**State: Normal Operation (Restored)**
Expected Logic: System permits initialization but prohibits active HV ramping.	**Expected Logic:** Hard system interlock. HV generation completely completely disabled.	**Expected Logic:** Safety loop closed. System fully armed for HV ramp testing.
PERMIT	✕ DISABLED	✓ ARMED

Gemini Notebook